

EDUCATION

Lovely Professional University
Bachelor of Technology (Computer Science & Engineering)
Jalandhar, India
July 2022 - May 2026

SKILLS SUMMARY

- **Programming:** Python, SQL
- **Machine Learning:** Scikit-learn, PyTorch, TensorFlow, Transformers, Model Evaluation
- **Analytics:** Statistics, Probability, Predictive Modeling
- **Generative AI:** LLMs, Prompt Engineering, Embeddings, Tokenisation, Attention, NLP
- **LLM Frameworks:** LangChain, LangGraph
- **RAG & Retrieval:** Vector Databases, FAISS, Qdrant, Semantic Search, Reranking, Retrieval Evaluation
- **ML Systems:** AWS, Docker, Git, Model Inference, Evaluation Frameworks, Monitoring, Deployment

WORK EXPERIENCE

- AI Intern | Know AI | [LINK](#) June 2026 - August 2026
- Built Generative AI and LLM applications using Python and PyTorch focused on model inference and evaluation.
 - Developed Python AI services and REST APIs integrating LLMs, tool/function calling, RAG, and automation.
 - Designed **RAG pipelines** using embeddings, vector databases, and semantic search for knowledge-grounded retrieval.
 - Developed AI/ML evaluation pipelines for model testing, error analysis, output validation, and reliability analysis.
 - Improved AI system reliability via evaluation, monitoring, debugging, latency optimization, and performance tuning.
- Data Engineer Intern | flickDONE | [LINK](#) June 2025 - August 2025
- Built scalable Python AI/ML data pipelines processing 50,000+ records for preprocessing and feature engineering.
 - Engineered automated **data preprocessing and validation workflows**, reducing manual verification by 70%.
 - Optimized AWS-based AI/ML pipelines, improving execution speed by 25% through performance tuning.
 - Developed reusable AI workflows for **document processing, LLM applications, and enterprise automation**.
 - Collaborated with engineering teams to improve **scalability, reliability, maintainability, and deployment** of ML platforms.

PROJECTS

- Ledgerly — Multi-Tenant SaaS Billing Platform | [LIVE](#) April 2026 - September 2026
- Built a multi-tenant SaaS platform using Next.js, TypeScript, NestJS, PostgreSQL, Prisma, and JWT.
 - Implemented tenant-isolated customer and invoice workflows with **RBAC and transactional data operations**.
 - Built **Stripe payment workflows and webhook processing** with idempotency and transactional updates.
 - Added **rate limiting, health checks, and secure error handling** for production-grade reliability.
 - Developed **174 automated tests across 21 suites** covering authentication, billing, payments, and tenant isolation.
- LangGraph Workflow Orchestrator | [LIVE](#) August 2025
- Built and deployed a stateful **Agentic AI platform** using Python, LangGraph, LangChain, FastAPI, and Gemini.
 - Designed multi-step LLM conversation workflows with state management, conditional routing, and error recovery.
 - Developed modular components for **task decomposition, tool calling, function calling, and agent coordination**.
 - Exposed AI workflows through FastAPI REST APIs, enabling scalable model-backed service deployment.
 - Improved workflow reliability through **checkpointing, testing, failure handling, and monitoring**.
- Production-Grade RAG Engine | [LIVE](#) June 2024 - August 2024
- Built an end-to-end production-grade RAG system using Python, FastAPI, LangChain, Qdrant, and AWS.
 - Developed document preprocessing, chunking, embeddings, indexing, retrieval, and response generation pipelines.
 - Implemented **semantic vector search with Qdrant** for low-latency, knowledge-grounded context retrieval.
 - Added reranking and retrieval evaluation to improve retrieval quality, context relevance, and AI response performance.
 - Improved AI service reliability through testing, monitoring, fallback handling, and error analysis.

CERTIFICATES

- Python, Data Science & ML (Cipherschools) | [CERTIFICATE](#) August 2024
- Built machine learning models using PyTorch and applied ML techniques for model development and evaluation.
 - Developed **Dockerized environments** supporting reproducible experimentation and deployment
- Generative AI with Large Language Models (Coursera) | [CERTIFICATE](#) March 2024
- Applied prompt engineering techniques for production LLM applications.
 - Built **retrieval workflows, AI output evaluation, and RAG pipeline** performance validation.